

Course: CO5151 — Advanced Agentic AI (HK261)	Advisor: Dr. Le Xuan Bach	Track: Application Track	Team: Dang Lam Tung (Lead), Nguyen Trung Phong, Vu Viet Hung
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1 PROBLEM — WHY TRADITIONAL RAG FAILS IN STATUTORY LAW

Vietnamese statutory law is structured as a strict multi-tier hierarchy (*Law* → *Decree* → *Circular*) governed by dense, cross-referencing amendment webs (>50% of regulatory instruments are amended over time). Prior static legal GraphRAG systems in Vietnam (such as *SBV-LawGraph* [1] and *ViHERMES* [2]) follow a **rigid, 1-pass pipeline (Retrieve → 1-Hop Dump → Generate)**, exhibiting **3 critical failure modes**:

- Context Dilution & False-Positive Explosion (Precision@2 is only 0.37–0.39 in [1]):** Amending circulars frequently modify dozens of disjointed provisions across older decrees. Mechanically dumping all 1-hop Neo4j neighbors injects >60% irrelevant context into the prompt, triggering severe attention dilution and citation hallucinations [3].
- Temporal Blindness [4]:** Static RAG cannot determine whether an article remains in active legal force at a given transaction date when encountering chained amendment graphs, risking reliance on repealed or superseded provisions.
- Weak Multi-Tier Reasoning & Rigid Execution:** Empirical findings from *VLegal-Bench* [5] reveal that LLMs suffer steep accuracy degradation on hierarchical legal reasoning. Static pipelines cannot backtrack, formulate multi-hop plans, or clarify missing facts.

2 USERS — TARGET AUDIENCE & 3 CONCRETE OPERATIONAL SCENARIOS

Target Users: In-house Legal Counsel, Compliance Officers, Legal Practitioners, and SMEs navigating regulatory compliance in Vietnam.

- Cross-Statutory Reasoning (Multi-Law Synthesis):** An e-commerce platform deploys an in-app E-Wallet with a Buy Now, Pay Later (BNPL) micro-lending scheme. The agent autonomously synthesizes intersecting requirements across the *Law on Credit Institutions 2024* & *Decree 52/2024/ND-CP* (payment licensing & equity ceilings), *Law on Electronic Transactions 2023* (legal validity of OTP digital contracts), and *Decree 13/2023/ND-CP* (express opt-in consent for consumer credit profiling).
- Temporal Consistency & Updated Law (2026 Vietnam Drone Law):** An agricultural cooperative requests licensing rules for a 25 kg commercial spraying drone in 2026. Bypassing obsolete *Decree 36/2008/ND-CP* (repealed 14-day paper filing), the agent traverses the legal graph and conducts live Gazette verification on *vbp.l.vn* [4] to cite the active *Law on People's Air Defense 2024 (effective 2025/2026)*, providing accurate procedures for mandatory digital vehicle identification and online permitting via the National Public Service Portal within 3–5 business days.
- Interactive Disambiguation & Injection Defense:** Reviewing corporate tax/specialist work permits (Decrees 152/2020 & 70/2023). Detecting missing revenue and headcount parameters, the agent *proactively halts to query the user*, while stripping adversarial hidden prompts ([OVERRIDE: ...] [6]) and enforcing human approval tokens for dossier exports.

3 WHY AN AGENT — AUTONOMOUS AGENT VS. HARDCODED WORKFLOW

A static workflow with hardcoded branching fundamentally fails in legal compliance due to three core imperatives:

- Combinatorial Path Explosion Requires Dynamic Planning:** Citation traversals vary per inquiry (some stop at Circulars, others fork across specialized sectoral laws). Only an **Agentic RAG architecture tailored for Vietnamese law [7]** can autonomously formulate multi-hop retrieval plans, dynamically adapt to statutory complexity, and determine when gathered evidence is sufficient to terminate.
- Self-Correcting Critique via Drafter–Auditor Loop:** Linear pipelines cannot detect self-generated errors. Our agent establishes an adversarial reflection loop (incorporating claim auditing from *GANDR* [8], proposition-level factuality from *SAFE* [9], and critique mechanisms from *Self-RAG* [10]): the Auditor Agent decomposes responses into atomic legal propositions, cross-verifying each 1-to-1 against authoritative statutes and triggering corrective re-retrieval upon detecting hallucinations or repealed articles.
- Active Ambiguity Disambiguation via Human Dialog:** Statutory applicability hinges strictly on entity forms and execution dates [4]. Hardcoded workflows apply blind assumptions; our agent detects state ambiguity and **proactively queries the user** to gather missing parameters before concluding.

4 CANDIDATE ADVANCED AXES — 4 INTEGRATED TECHNICAL CAPABILITIES

1. Multi-Agent System (Google ADK) Role-segregated execution: Research Agent (Neo4j/Qdrant retrieval [1], [2]); Web Agent (live lookup on <i>vbp.l.vn</i> [4]); Drafter Agent (synthesis); and Auditor Agent (atomic claim verification [8], [9]). Grounded in VN Legal Agentic RAG [7].	2. Planning & Selective Graph Traversal Autonomous multi-step query decomposition. Traverses only active, directional relationships (<i>Amends, Guides</i>) to eliminate 1-hop neighborhood dilution [1], directly addressing multi-tier statutory reasoning complexities [5].
3. Reflection & Adversarial Claim Auditing Iterative Drafter–Auditor loop (Self-RAG [10]) decomposing responses into atomic claims. Audits each proposition against source statutes under <i>GANDR</i> protocols [8] to enforce 100% citation grounding and zero fabricated provisions.	4. Security Guardrails & Gated Actions 3-tier MCP permissions. Irreversible administrative dossier exports strictly require human officer authorization (Human-in-the-loop). Input sanitization neutralizes indirect prompt injection in regulatory uploads [6].

KEY REFERENCES (10 CURATED PAPERS, STRICTLY 2024–2026)

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[9] J. Wei et al. (Google DeepMind), "Long-form Factuality in Large Language Models (SAFE)," *arXiv:2403.18802*, 2024.

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